Appendix: Technical Details and Decision Log

Project #1: What Do Negative and Positive Amazon Reviews Talk About? | Amir Ali

1. **Data source:** Amazon Reviews dataset from Kaggle (dongrelaxman/amazon-reviews-dataset). 21,214 reviews received. Each row is one review.
2. **Star rating and sentiment:** Stars were taken from the Rating column. Negative = 1–2 stars, Neutral = 3 stars, Positive = 4–5 stars.
3. **Text used:** The review title and review text were joined into one column (Review Combined), so each review is one document in one cell.
4. **Cleaning:** Empty placeholder text ("Review text not found") was removed, producing the Review Clean column used for text analysis.
5. **Years kept:** Only 2018–2024 were kept because each of those years had at least 1,000 reviews. Positive reviews became much less common over the years, so older years were left out. Final data: 18,130 reviews.
6. **Text Explorer:** Similar word forms were combined (stemming). Common filler words and rare terms were added to the stop word list, and the word "amazon" was removed because it appears in nearly every review. A minimum term frequency was used to drop very rare words.
7. **Topic Analysis:** 10 topics (rotated SVD, 100 singular vectors, TF-IDF weighting). Topics 3 and 5 contained only symbols and foreign-language text and were ignored. Topic scores are centered: 0 = average, positive = stronger match than average, negative = weaker match than average.
8. **Review length:** Negative reviews were much longer than positive ones, which could distort the comparison. Topic and cluster comparisons therefore use reviews of similar length (150–450 characters, 7,455 reviews).
9. **Cluster Analysis:** K Means with 4 clusters, chosen using the Cubic Clustering Criterion. Cluster names are our own description after reading review titles and ratings in each cluster.
10. **Limits:** The data are reviews of Amazon as a company, written by people who chose to post. 78% of reviews are negative, so results describe reviewers, not all customers.